Week 01 teacher guide

Fractions and equal parts

PowerPoint slides 1-106 | Student workbook pp. 3-13

The whole stays the same when fractions are compared. Multiplication can take part of an amount. Division can count fractional groups or split a fraction into shares.

The seven days

Day 001 · pp. 3-5 · Saturday · 20 questions,including 2 challenges
Fractions on one number line

Day 002 · pp. 6-8 · Sunday · 20 questions,including 2 challenges
Fraction operations with models

Day 003 · p. 9 · Monday · 20 questions,including 2 challenges
Fractions and equal parts

Day 004 · p. 10 · Tuesday · 20 questions,including 2 challenges
Fractions and equal parts

Day 005 · p. 11 · Wednesday · 25 questions,including 2 challenges
Fractions and equal parts

Day 006 · p. 12 · Thursday · 20 questions,including 2 challenges
Fractions and equal parts

Day 007 · p. 13 · Friday · 20 questions,including 2 challenges
Weekly check

Use one day at a time

Each weekend lesson uses a three-page packet; each weekday uses one question page. Work on separate plain paper, labeling the Day number and item IDs. Weekend lessons and ordinary weekdays contain 20 questions including C1 and C2; Wednesday contains 25. Allow 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Split a demanding day into two sittings if needed.

The weekend packet includes models and worked examples, or a scheduled assessment alternative. Read a worked example after an attempt. Keep answer-review slides out of view until the corresponding work is complete. Weekly answer review is optional after Friday and does not add to the 80-minute weekend lesson.

Built-in review

Day 003: Fractions: compare and combine (9, 10, 11, 12)

Day 004: Fraction operations (9, 10, 11, 12)

Day 005: Weekly puzzle (13)

Day 006: Multiplication and division facts (9, 10, 11, 12)

On Friday, record routine accuracy out of 18 separately from the two challenge explanations. For each challenge, note whether the learner connected the steps, used a valid model or calculation, and justified the conclusion. Revisit a missed model, then check with a fresh item later.

Assessment alternatives: Diagnostic: fact snapshot (p. 5); Diagnostic: computation and reasoning (p. 4). Replace ordinary practice or use a separate assessment sitting; do not add these on top of the assigned day.

Saturday. Fractions on one number line

Day 001 · pp. 3-5 | Slides 4-31

Goal: I can compare fractions and decimals by placing them on the same number line.

Check first: Name numerator and denominator. Divide a whole into equal parts.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 4-5 · Orient
Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 6-7 · Explore · P1, P2, P3
Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 8-12 · Model
Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 13-15 · Explore · P4, P5, P6, P7, P8, F
Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 16 · Discuss an error · D
Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 17-22 · Guided practice · X1, X2, X3, X4, X5, X6
Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 23-24 · Deepening
Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 25-28 · Challenges · C1, C2
Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 29 · Summary · S
Ask for one sentence or annotated example in the learner's own words. Listen for the main relationship.

Slides 30 · Exit · E
Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 31 · Tutor check
After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Saturday. Discussion and support

Investigation T: One fraction, two different wholes

Watch and discuss; this is a projected investigation, not an extra assigned question.

Two strips are 1 meter and 2 meters long. Shade $\frac{3}{4}$ of each. Compare the shaded lengths. Then explain when a fraction comparison alone is enough.

1. Label each whole before shading.
2. Partition both wholes into 4 equal parts.
3. Shade 3 parts on each and measure the shaded lengths.
4. State the condition needed for comparing fractions alone.

Ask: Can equal fractions name unequal lengths? What must stay fixed when we compare fractions?

If stuck: Align the strips at zero and label each whole length.

Expected: The shaded lengths are $\frac{3}{4}$ meter and 1 $\frac{1}{2}$ meters. The second is twice as long; fractions alone compare amounts only when the wholes match.

C1 · Challenge discussion

Ask: Why must both positions use the same whole?

If stuck: Rename both fractions with denominator 60.

Expected: Ari; $\frac{1}{60}$ of the trail

C2 · Challenge discussion

Ask: Can someone have less liquid but need less to reach the same target?

If stuck: Draw each starting amount and the common target on one number line.

Expected: Lina; Lina needs $\frac{1}{8}$ liter and Omar needs $\frac{3}{20}$ liter

What to notice

Uses equal-length wholes. → Ask the student to mark 0 and 1 on both strips.

Uses a benchmark or common denominator. → Accept either method and ask where the fractions lie.

Common error: Places $\frac{5}{4}$ between 0 and 1. Repair: Build four fourths, then add one fourth.

Sunday. Fraction operations with models

Day 002 · pp. 6-8 | Slides 32-59

Goal: I can use equal parts to explain fraction multiplication and two kinds of division.

Check first: Equivalent fractions and equal-sized parts.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 32-33 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 34-35 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 36-37 · Explore · P1, P2

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 38 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 39-41 · Explore · P3, P4, P5, P6, P7

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 42-44 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 45 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 46 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 47 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 48 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 49 · Explore · M1, M2, M3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 50-51 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 52-55 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 56 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 57 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 58 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 59 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Sunday. Discussion and support

Investigation T: A multiplier changes size

Watch and discuss; this is a projected investigation, not an extra assigned question.

Predict whether $\frac{3}{4} \times 8$, 1×8, and $\frac{5}{4} \times 8$ are below, equal to, or above 8. Use one model to explain, then compute. Would the same strict comparisons work if the starting amount were 0?

1. Compare each multiplier with 1 before calculating.
2. Split 8 into 4 equal groups of 2.
3. Take 3, 4, and 5 groups to get 6, 8, and 10.
4. Test the claim at 0 and state its positive-amount condition.

Ask: Why does comparing the factor with 1 predict the size for a positive amount? What changes when the starting amount is 0?

If stuck: Split 8 into 4 equal groups, then choose 3, 4, or 5 of those groups.

Expected: $6 < 8$; $8 = 8$; $10 > 8$. With a starting amount of 0, all three products are 0, so the strict smaller/larger rule requires a positive starting amount.

C1 · Challenge discussion

Ask: Where do you see the two batches and the six cups in one picture?

If stuck: Draw two strips, each with three fourths shaded.

Expected: 6 cups

C2 · Challenge discussion

Ask: Why do we compare area with area rather than a side length?

If stuck: First label the rug length and width; multiply to get its area.

Expected: The rug; $\frac{1}{5}$ square meter more

What to notice

Counts overlapping equal parts in the area model. → Ask what fraction of the entire unit square is shaded twice.

Predicts shrinking for a factor below one. → Ask for an estimate before calculation.

Common error: Says multiplication always increases an amount. Repair: Take half of a familiar amount.

Slide-order index

Follow this numerical order. The session plans explain how to teach each phase; full scripts remain in PowerPoint speaker notes. Item IDs match the student workbook.

Orientation

1. Double Advanced Math Program
Start here · Workbook

2. How a session runs
Watch and discuss

3. The twenty sessions
Watch and discuss

Saturday

4. Week 1, Saturday
Watch and discuss

5. Today
Watch and discuss · The baseline diagnostic is a separate assessment option.

6. Same denominator
Day 001 · Workbook p. 3 · P1

7. Same numerator
Day 001 · Workbook p. 3 · P2, P3

8. The model
Watch and discuss

9. Worked example · 1 of 4
Watch and discuss · worked example

10. Worked example · 2 of 4
Watch and discuss · worked example

11. Worked example · 3 of 4
Watch and discuss · worked example

12. Worked example · 4 of 4
Watch and discuss · worked example

13. Unlike denominators and numerators
Day 001 · Workbook p. 3 · P4, P5

14. Fraction versus decimal
Day 001 · Workbook p. 3 · P6, P7

15. Fraction as division
Day 001 · Workbook p. 3 · P8, F

16. A claim to check
Day 001 · Workbook p. 3 · D

17. Guided practice · X1
Day 001 · Workbook p. 3 · X1

18. Guided practice · X2
Day 001 · Workbook p. 3 · X2

19. Guided practice · X3
Day 001 · Workbook p. 3 · X3

20. Guided practice · X4
Day 001 · Workbook p. 3 · X4

21. Guided practice · X5
Day 001 · Workbook p. 3 · X5

22. Guided practice · X6
Day 001 · Workbook p. 3 · X6

23. One fraction, two different wholes
Watch and discuss · draw on separate paper

24. One fraction, two different wholes
Watch and discuss · draw on separate paper

25. Challenge C1
Day 001 · Workbook p. 5 · C1

26. Challenge C1 · Discuss
Day 001 · Workbook p. 5 · C1

27. Challenge C2
Day 001 · Workbook p. 5 · C2

28. Challenge C2 · Discuss
Day 001 · Workbook p. 5 · C2

29. In your own words
Day 001 · Workbook p. 3 · S

30. Exit ticket
Day 001 · Workbook p. 3 · E

31. Tutor reference
Tutor only;hidden in slide show

Sunday

32. Week 1, Sunday
Watch and discuss

33. Today
Watch and discuss

34. Recall
Day 002 · Workbook p. 6 · R1, R2, R3, R4

35. Recall: answers
Day 002 · Workbook p. 6 · R1, R2, R3, R4

36. Whole number times fraction
Day 002 · Workbook p. 6 · P1

37. Fraction times fraction
Day 002 · Workbook p. 6 · P2

38. The model
Watch and discuss

39. Fraction times mixed number
Day 002 · Workbook p. 6 · P3, P4

40. Whole number divided by unit fraction
Day 002 · Workbook p. 7 · P5, P6

41. Contrast counting groups and sharing
Day 002 · Workbook p. 7 · P7

42. Worked example · 1 of 3
Watch and discuss · worked example

43. Worked example · 2 of 3
Watch and discuss · worked example

44. Worked example · 3 of 3
Watch and discuss · worked example

45. Your turn
Day 002 · Workbook p. 7 · F

46. Two ways to see it
Watch and discuss · compare the same example

47. A claim to check
Day 002 · Workbook p. 7 · D

48. Words and meaning
Watch and discuss

49. Mixed review
Day 002 · Workbook p. 7 · M1, M2, M3

50. A multiplier changes size
Watch and discuss · draw on separate paper

51. A multiplier changes size
Watch and discuss · draw on separate paper

52. Challenge C1
Day 002 · Workbook p. 8 · C1

53. Challenge C1 · Discuss
Day 002 · Workbook p. 8 · C1

54. Challenge C2
Day 002 · Workbook p. 8 · C2

55. Challenge C2 · Discuss
Day 002 · Workbook p. 8 · C2

56. In your own words
Day 002 · Workbook p. 8 · S

57. Exit ticket
Day 002 · Workbook p. 8 · E

58. The week ahead
Days 003-007 · Workbook pp. 9-13

59. Tutor reference
Tutor only;hidden in slide show

Answer review

60. Week 1 answers
Next lesson: slide 107. Use this review section after Friday.

61. Fractions on one number line · 1 of 6
Day 001 · Workbook p. 3 · Items P1, P2, P3, P4

62. Fractions on one number line · 2 of 6
Day 001 · Workbook p. 3 · Items P5, P6, P7, P8

63. Fractions on one number line · 3 of 6
Day 001 · Workbook p. 3 · Items F, D, X1

64. Fractions on one number line · 4 of 6
Day 001 · Workbook p. 3 · Items X2, X3, X4, X5

65. Fractions on one number line · 5 of 6
Day 001 · Workbook pp. 3, 5 · Items X6, S, E, C1

66. Fractions on one number line · 6 of 6
Day 001 · Workbook p. 5 · Items C2

67. Diagnostic: fact snapshot · 1 of 2
Day 001 · Workbook p. 5 · Items 1-18

68. Diagnostic: fact snapshot · 2 of 2
Day 001 · Workbook p. 5 · Items 19-30

69. Diagnostic: computation and reasoning · 1 of 6
Day 001 · Workbook p. 4 · Items 1, 2, 3, 4, 5

70. Diagnostic: computation and reasoning · 2 of 6
Day 001 · Workbook p. 4 · Items 6, 7, 8, 9

71. Diagnostic: computation and reasoning · 3 of 6
Day 001 · Workbook p. 4 · Items 10, 11, 12, 13

72. Diagnostic: computation and reasoning · 4 of 6
Day 001 · Workbook p. 4 · Items 14, 15, 16, 17, 18

73. Diagnostic: computation and reasoning · 5 of 6
Day 001 · Workbook p. 4 · Items 19, 20, 21, 22

74. Diagnostic: computation and reasoning · 6 of 6
Day 001 · Workbook p. 4 · Items 23, 24, 25

75. Fraction operations with models · 1 of 5
Day 002 · Workbook p. 6 · Items R1, R2, R3, R4

76. Fraction operations with models · 2 of 5
Day 002 · Workbook p. 6 · Items P1, P2, P3, P4

77. Fraction operations with models · 3 of 5
Day 002 · Workbook p. 7 · Items P5, P6, P7, F

78. Fraction operations with models · 4 of 5
Day 002 · Workbook p. 7 · Items D, M1, M2, M3

79. Fraction operations with models · 5 of 5
Day 002 · Workbook p. 8 · Items S, E, C1, C2

80. Fractions and equal parts · 1 of 6
Day 003 · Workbook p. 9 · Items 1, 2, 3, 4, 5

81. Fractions and equal parts · 2 of 6
Day 003 · Workbook p. 9 · Items 6, 7, 8, 9

82. Fractions and equal parts · 3 of 6
Day 003 · Workbook p. 9 · Items 10, 11, 12

83. Fractions and equal parts · 4 of 6
Day 003 · Workbook p. 9 · Items 13, 14, 15

84. Fractions and equal parts · 5 of 6
Day 003 · Workbook p. 9 · Items 16, 17, 18, C1

85. Fractions and equal parts · 6 of 6
Day 003 · Workbook p. 9 · Items C2

86. Fractions and equal parts · 1 of 5
Day 004 · Workbook p. 10 · Items 1, 2, 3, 4, 5

87. Fractions and equal parts · 2 of 5
Day 004 · Workbook p. 10 · Items 6, 7, 8, 9

88. Fractions and equal parts · 3 of 5
Day 004 · Workbook p. 10 · Items 10, 11, 12, 13

89. Fractions and equal parts · 4 of 5
Day 004 · Workbook p. 10 · Items 14, 15, 16, 17

90. Fractions and equal parts · 5 of 5
Day 004 · Workbook p. 10 · Items 18, C1, C2

91. Fractions and equal parts · 1 of 6
Day 005 · Workbook p. 11 · Items 1, 2, 3, 4, 5

92. Fractions and equal parts · 2 of 6
Day 005 · Workbook p. 11 · Items 6, 7, 8, 9

93. Fractions and equal parts · 3 of 6
Day 005 · Workbook p. 11 · Items 10, 11, 12, 13

94. Fractions and equal parts · 4 of 6
Day 005 · Workbook p. 11 · Items 14, 15, 16, 17

95. Fractions and equal parts · 5 of 6
Day 005 · Workbook p. 11 · Items 18, 19, 20, 21

96. Fractions and equal parts · 6 of 6
Day 005 · Workbook p. 11 · Items 22, 23, C1, C2

97. Fractions and equal parts · 1 of 5
Day 006 · Workbook p. 12 · Items 1, 2, 3, 4, 5

98. Fractions and equal parts · 2 of 5
Day 006 · Workbook p. 12 · Items 6, 7, 8, 9

99. Fractions and equal parts · 3 of 5
Day 006 · Workbook p. 12 · Items 10, 11, 12, 13, 14

100. Fractions and equal parts · 4 of 5
Day 006 · Workbook p. 12 · Items 15, 16, 17, 18

101. Fractions and equal parts · 5 of 5
Day 006 · Workbook p. 12 · Items C1, C2

102. Weekly check · 1 of 5
Day 007 · Workbook p. 13 · Items 1, 2, 3, 4

103. Weekly check · 2 of 5
Day 007 · Workbook p. 13 · Items 5, 6, 7, 8, 9

104. Weekly check · 3 of 5
Day 007 · Workbook p. 13 · Items 10, 11, 12

105. Weekly check · 4 of 5
Day 007 · Workbook p. 13 · Items 13, 14, 15, 16

106. Weekly check · 5 of 5
Day 007 · Workbook p. 13 · Items 17, 18, C1, C2

Complete weekly answers

Match Day,item ID,and page before checking. Equivalent correct forms and valid alternative methods are welcome. Computational keys are compact; challenges and conceptual items include reasoning. No challenge is complete with an unsupported answer alone.

Day 001 · Fractions on one number line

Workbook pp. 3-5 · assigned set

P1 · F-COMP
$\frac{3}{7} < \frac{5}{7}$

Reason: Both use sevenths. Three parts are fewer than five.

P2 · F-COMP
$\frac{4}{5} > \frac{4}{9}$

Reason: Fifths are larger than ninths, so four fifths cover more.

P3 · F-COMP
Less than $\frac{1}{2}$

P4 · F-COMP
$\frac{3}{5} > \frac{7}{12}$

Reason: $\frac{3}{5} = \frac{36}{60}$; $\frac{7}{12} = \frac{35}{60}$. Compare 36 and 35.

P5 · F-LINE
Left to right: 0.5, $\frac{5}{4}$, $1\ \frac{3}{4}$

P6 · F-COMP
$\frac{7}{10} > 0.68$

Reason: $\frac{7}{10} = 0.70$, which is 2 hundredths more than 0.68.

P7 · F-LINE
One fourth of the way from 2 to 4

P8 · F-QUOT
$\frac{3}{4}$ of a muffin

F · F-COMP
10 twelfths; $\frac{5}{6} > \frac{3}{4}$

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · F-COMP
$\frac{4}{5} > \frac{4}{9}$. Nine equal pieces are smaller than five equal pieces of the same whole.

Reason: She compares denominator counts without comparing piece sizes. Align equal-length strips. Cover four fifths and four ninths.

X1 · F-COMP
$\frac{6}{14} < \frac{8}{14}$

Reason: The parts have equal size. The second fraction counts two more parts.

X2 · F-COMP
$\frac{3}{5}$

Reason: With the same whole and numerator, fewer equal parts make each part larger.

X3 · F-COMP
$\frac{7}{8} < \frac{8}{9}$

Reason: Cross-products are 63 and 64; both fractions can use denominator 72.

X4 · F-COMP
$\frac{3}{4} > 0.74$

Reason: $\frac{3}{4} = 0.75$. Compare hundredths or thousandths.

X5 · F-LINE
$1\ \frac{3}{8}$

X6 · F-COMP
One answer: $\frac{5}{8}$

S · EXPLAIN
Answers vary. A fraction names equal parts of one whole. The size of that whole stays fixed.

Reason: Accept an accurate explanation supported by a matching example.

E · F-COMP
$\frac{7}{10}$, 0.72, $\frac{3}{4}$

Reason: Rename as 0.70, 0.72, 0.75. Their positions increase to the right.

C1 · F-COMP
Ari; $\frac{1}{60}$ of the trail

Reason: Rename in sixtieths: $\frac{3}{5} = \frac{36}{60}$ and $\frac{7}{12} = \frac{35}{60}$. Ari is one sixtieth farther. Use the same whole trail for both positions.

C2 · F-COMP
Lina; Lina needs $\frac{1}{8}$ liter and Omar needs $\frac{3}{20}$ liter

Reason: Use fortieths: $\frac{3}{4} = \frac{30}{40}$, $\frac{5}{8} = \frac{25}{40}$, and $0.6 = \frac{24}{40}$. The gaps are $\frac{5}{40}$ and $\frac{6}{40}$; Lina needs less.

Day 001 · Diagnostic: fact snapshot

Workbook p. 5 · assessment alternative

1 · N-FACT
20

2 · N-FACT
18

3 · N-FACT
77

4 · N-FACT
56

5 · N-FACT
64

6 · N-FACT
12

7 · N-FACT
42

8 · N-FACT
35

9 · N-FACT
88

10 · N-FACT
22

11 · N-FACT
63

12 · N-FACT
54

13 · N-FACT
16

14 · N-FACT
120

15 · N-FACT
20

16 · N-FACT
28

17 · N-FACT
15

18 · N-FACT
8

19 · N-FACT
48

20 · N-FACT
60

21 · N-FACT
22

22 · N-FACT
10

23 · N-FACT
66

24 · N-FACT
18

25 · N-FACT
56

26 · N-FACT
72

27 · N-FACT
70

28 · N-FACT
48

29 · N-FACT
14

30 · N-FACT
84

Day 001 · Diagnostic: computation and reasoning

Workbook p. 4 · assessment alternative

1 · N-MULT
9798

2 · N-DIV
228

3 · N-DIV
91

4 · D-OPS
18.45

5 · D-OPS
4.63

6 · D-OPS
0.28

7 · D-OPS
7

8 · F-MULT
$\frac{1}{2}$

9 · F-ADD
$1\frac{1}{12}$

10 · F-DIV
12

11 · F-DIV
$\frac{1}{15}$

12 · F-MULT
7

13 · F-COMP
$\frac{4}{9} < \frac{1}{2}$

14 · F-COMP
Left to right: $\frac{5}{8}$, $\frac{3}{4}$, $\frac{4}{3}$, $1\frac{1}{2}$

15 · D-COMP
0.506, 0.56, 0.6

16 · R-RATIO
10

17 · R-PERCENT
16

18 · E-EXPR
27

19 · N-SIGNED
6°C

20 · G-AREA
$\frac{1}{2}$ m²

21 · G-VOL
60 cm³

22 · N-COORD
3 units

23 · F-SCALE
6 is two of three equal shares of 9. A factor between 0 and 1 scales down.

24 · P-LOGIC
20, 27

25 · P-LOGIC
4, 6, 8

Day 002 · Fraction operations with models

Workbook pp. 6-8 · assigned set

R1 · G5-FACT
56

R2 · G5-FACT
8

R3 · F-ADD
$\frac{3}{4}$

R4 · D-ADD
0.85

P1 · F-MULT
$3\frac{3}{4}$ cups

P2 · F-MULT
$\frac{2}{5}$ square meter

P3 · F-MULT
1 cup

P4 · F-SCALE
$6 < 8$; $10 > 8$

Reason: Three fourths takes less than all eight. Five fourths takes more than all eight.

P5 · F-DIV
12 pieces

P6 · F-DIV
$\frac{1}{12}$ liter

P7 · F-DIV
10 groups; $\frac{1}{10}$ in each share. Stories vary.

F · F-DIV
$\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · F-DIV

Twelve quarter-meter pieces fit in 3 meters.

Reason: He treats division by a fraction as multiplication by that fraction. Make one meter from four quarter-meter pieces, then repeat three times.

M1 · G5-FACT

52

M2 · G5-FACT

8

M3 · F-ADD

$\frac{1}{2}$

S · EXPLAIN

Answers vary. Multiplication can take part of an amount. Division can count groups or find the size of each share.

Reason: Accept an accurate explanation supported by a matching example.

E · F-DIV

10 shelves

Reason: Each meter contains two half-meters, so five meters contain ten.

C1 · F-MULT

6 cups

Reason: First find total smoothie: $2 \times \frac{3}{4} = \frac{3}{2}$ liters. A whole liter holds four quarter-liter cups, so $\frac{3}{2}$ liters holds 6 cups. A strip of six quarters shows both operations.

C2 · F-MULT

The rug; $\frac{1}{5}$ square meter more

Reason: Rename $1\ \frac{1}{2}$ as $\frac{3}{2}$. The rug area is $\frac{3}{2} \times \frac{2}{3} = 1$ square meter. Subtract $\frac{4}{5}$ to find the difference $\frac{1}{5}$ square meter.

Day 003 · Fractions and equal parts

Workbook p. 9 · assigned set

1 · N-FACT

9

2 · N-FACT

21

3 · N-FACT

12

4 · N-FACT

27

5 · N-FACT

12

6 · N-FACT

40

7 · N-FACT

16

8 · F-COMP

$\frac{1}{3} > \frac{1}{4}$

Reason: With the same numerator, the fraction with larger pieces is larger.

9 · F-COMP · review

$\frac{2}{9} < \frac{7}{9}$

Reason: With ninths fixed, compare two parts and seven parts.

10 · F-COMP · review

$\frac{2}{4} > \frac{2}{5}$

Reason: With the same numerator, the fraction with larger pieces is larger.

11 · F-COMP · review

$\frac{2}{5}$, 0.45, $\frac{1}{2}$

12 · F-COMP · review

$\frac{5}{8} > \frac{3}{5}$

Reason: Rename as twenty-five fortieths and twenty-four fortieths.

13 · F-COMP

$\frac{4}{5} > \frac{4}{9}$. Nine equal pieces are smaller than five equal pieces of the same whole.

14 · F-COMP

$\frac{9}{14} < \frac{11}{14}$

Reason: The parts have equal size. The second fraction counts two more parts.

15 · F-COMP

$\frac{2}{6}$

Reason: With the same whole and numerator, fewer equal parts make each part larger.

16 · F-COMP

$\frac{3}{7} > \frac{5}{12}$

Reason: Cross-products are 36 and 35; both fractions can use denominator 84.

17 · F-COMP
$\frac{1}{4} > 0.24$

Reason: $\frac{1}{4} = 0.25$. Compare hundredths or thousandths.

18 · F-LINE
$1\ \frac{2}{3}$

C1 · F-ADD
Sam; $\frac{1}{24}$ mile

Reason: Jade walks $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$. Compare $\frac{15}{24}$ with $\frac{16}{24}$; Sam walks $\frac{1}{24}$ mile farther.

C2 · F-SUB
Above half by $\frac{1}{12}$ of the jar

Reason: Both fractions refer to full capacity. $\frac{5}{6} - \frac{1}{4} = \frac{10}{12} - \frac{3}{12} = \frac{7}{12}$. Half is $\frac{6}{12}$, so the remainder is $\frac{1}{12}$ above half.

Day 004 · Fractions and equal parts

Workbook p. 10 · assigned set

1 · F-EQUIV
0.5; 50%

2 · F-EQUIV
0.25; 25%

3 · F-EQUIV
0.75; 75%

4 · F-EQUIV
0.125; 12.5%

5 · N-FACT
15

6 · N-FACT
12

7 · N-FACT
12

8 · F-DIV
20; $\frac{1}{20}$

9 · F-MULT · review
2

10 · F-MULT · review
$\frac{4}{15}$

11 · F-DIV · review
6 pieces

12 · F-DIV · review
$\frac{1}{6}$ liter

13 · F-DIV
Twelve quarter-meter pieces fit in 3 meters.

14 · F-MULT
$\frac{5}{2}$ kilograms

15 · F-MULT
$\frac{5}{12}$

16 · F-MULT
$\frac{1}{4}$ liters

17 · F-DIV1
56

18 · F-DIV1
$\frac{1}{9}$ liter

C1 · F-MULT
Enough; $\frac{1}{2}$ meter left

Reason: Four packages use $4 \times \frac{5}{8} = \frac{20}{8} = \frac{5}{2}$ meters. $3 - \frac{5}{2} = \frac{1}{2}$ meter remains.

C2 · F-DIV1
The $\frac{1}{10}$-liter share; larger by $\frac{1}{60}$ liter

Reason: The first share is $\frac{1}{3} \div 4 = \frac{1}{12}$ liter. Compare $\frac{1}{10}$ and $\frac{1}{12}$ using sixtieths: $\frac{6}{60} - \frac{5}{60} = \frac{1}{60}$.

Day 005 · Fractions and equal parts

Workbook p. 11 · assigned set

1 · N-SCALE
32

2 · N-SCALE
3

3 · N-SCALE
7

4 · N-SCALE
9

5 · F-LINE
At $1\ \frac{3}{4}$, the seventh quarter-step from 0

6 · F-DIV
$\frac{1}{18}$; stories vary

7 · N-FACT
24

8 · F-ADD
$\frac{1}{3}$

9 · N-SCALE
300

10 · N-FACT
6

11 · N-FACT
36

12 · G-VOL
108 cm³

13 · PUZZLE · review
$\frac{5}{8}$

Reason: Half is four eighths and three fourths is six eighths. Only five eighths lies strictly between them.

14 · F-COMP
$\frac{2}{11} < \frac{4}{11}$

Reason: The parts have equal size. The second fraction counts two more parts.

15 · F-COMP
$\frac{2}{5}$

Reason: With the same whole and numerator, fewer equal parts make each part larger.

16 · F-COMP
$\frac{5}{12} < \frac{4}{9}$

Reason: Cross-products are 45 and 48; both fractions can use denominator 108.

17 · F-COMP
$\frac{1}{4} > 0.24$

Reason: $\frac{1}{4} = 0.25$. Compare hundredths or thousandths.

18 · F-LINE
$1\frac{2}{3}$

19 · F-COMP
One answer: $\frac{11}{20}$

20 · F-EQUIV
5

Reason: Multiply numerator and denominator by 5; the value stays unchanged.

21 · F-ADD
$\frac{11}{15}$

22 · F-SUB
$\frac{1}{2}$

Reason: Use denominator 18; subtract 6 from 15, then simplify. Add $\frac{1}{3}$ to check.

23 · N-FACT
10

C1 · F-COMP
n = 7 or 8; $\frac{7}{12} < \frac{8}{12}$

Reason: Half is $\frac{6}{12}$ and three fourths is $\frac{9}{12}$. The whole numerators strictly between 6 and 9 are 7 and 8. Equal denominators make the order immediate.

C2 · F-MULT
6 cups

Reason: Three pitchers hold $3 \times \frac{2}{3} = 2$ liters. Serving half a liter leaves $\frac{3}{2}$ liters. Six quarter-liter cups hold that amount.

Day 006 · Fractions and equal parts

Workbook p. 12 · assigned set

1 · N-EST
300

2 · N-EST
4

3 · N-EST
About 100

4 · N-EST
About 20

5 · F-LINE
At $1\frac{3}{4}$, the seventh quarter-step from 0

6 · F-DIV
$\frac{1}{18}$; stories vary

7 · F-COMP
$\frac{5}{8} > \frac{3}{5}$

Reason: Rename as twenty-five fortieths and twenty-four fortieths.

8 · G-VOL
30 cm³

9 · N-FACT · review
48

10 · N-FACT · review
8

11 · N-FACT · review
63

12 · N-FACT · review
9

13 · F-SCALE
Smaller; 6

Reason: Two thirds takes less than all nine.

14 · F-DIV
Twelve quarter-meter pieces fit in 3 meters.

15 · F-MULT
$\frac{14}{5}$ kilograms

16 · F-MULT
$\frac{2}{5}$

17 · F-MULT
$\frac{4}{5}$ liters

18 · F-DIV1
28

C1 · F-MULT
Garden: $\frac{3}{2}$ square meters; herbs: $\frac{3}{4}$ square meter

Reason: Multiply $\frac{5}{2} \times \frac{3}{5} = \frac{3}{2}$ for the full area. Half of that area is $\frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$. The second half must have the same area.

C2 · F-COMP
Kai; $\frac{1}{40}$ meter more

Reason: Kai has $\frac{7}{8} - \frac{2}{8} = \frac{5}{8} = 0.625$ meter. Ren has 0.6 meter = $\frac{3}{5}$. The difference is $\frac{25}{40} - \frac{24}{40} = \frac{1}{40}$ meter.

Day 007 · Weekly check

Workbook p. 13 · assigned set

1 · F-LINE
At $1\frac{3}{4}$, the seventh quarter-step from 0

2 · F-DIV
$\frac{1}{18}$; stories vary

3 · F-COMP
$\frac{5}{8} > \frac{3}{5}$

Reason: Rename as twenty-five fortieths and twenty-four fortieths.

4 · F-ADD
$\frac{5}{6}$

5 · G-VOL
54 cm³

6 · N-FACT
12

7 · D-OPS
6.75

8 · N-SCALE
5

9 · N-FACT
20

10 · F-COMP
$\frac{4}{5} > \frac{4}{9}$. Nine equal pieces are smaller than five equal pieces of the same whole.

11 · F-COMP
$\frac{8}{11} < \frac{10}{11}$

Reason: The parts have equal size. The second fraction counts two more parts.

12 · F-COMP
$\frac{2}{4}$

Reason: With the same whole and numerator, fewer equal parts make each part larger.

13 · F-COMP
$\frac{2}{3} > \frac{7}{12}$

Reason: Cross-products are 24 and 21; both fractions can use denominator 36.

14 · F-COMP

$\frac{3}{8} > 0.365$

Reason: $\frac{3}{8} = 0.375$. Compare hundredths or thousandths.

15 · F-LINE

$1\frac{2}{3}$

16 · F-COMP

One answer: $\frac{11}{20}$

17 · F-EQUIV

18

Reason: Multiply numerator and denominator by 3; the value stays unchanged.

18 · F-ADD

$\frac{7}{12}$

C1 · F-MULT

$\frac{1}{2}$ cup remains; less than one batch by $\frac{1}{6}$ cup

Reason: Three batches use 2 cups. The bag leaves $\frac{1}{2}$ cup. Compare $\frac{1}{2}$ and $\frac{2}{3}$: the remainder is $\frac{1}{6}$ cup short of another batch.

C2 · F-SCALE

The products are 8 and 9; her numerical conclusion is true, but the reasoning overlooks the different wholes.

Reason: $\frac{4}{5} \times 10 = 8$ and $\frac{3}{4} \times 12 = 9$. Comparing fractions alone is not enough when they act on different whole amounts. Draw ten-unit and twelve-unit bars to show the wholes.